Manasvi Agarwal

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SUMMARY

New grad Software Engineer with experience building scalable research infrastructure and REST APIs using Flask, Docker, and Google Cloud. Led the development of image privacy and classification features at a medical AI startup. Skilled at collaborating cross-functionally with researchers, product teams, and engineers to ship robust, production-grade systems. Passionate about accelerating ML workflows and enabling cutting-edge research through thoughtful tooling and backend reliability.

SKILLS

- **Programming Languages:** Python, Java, C++, JavaScript, SQL, R
- **ML/NLP:** scikit-learn, Naive Bayes, PyTorch (basic), Pandas, NumPy
- **Frontend:** React.js, HTML, CSS, TypeScript (basic), DOM manipulation
- **Backend & Tools:** Flask, REST APIs, Git, Docker, Linux, Google Cloud Platform, Google Colab

WORK EXPERIENCE

VEYTEL, INC — Remote

(A medical AI startup based in Pennsylvania specializing in advanced imaging solutions for early detection of melanoma and assessment of respiratory illnesses)

Assistant Computer Scientist — Jul 2025 – Present

- Developed and maintained Flask-based REST APIs to support medical image analysis features for a clinical AI web platform.
- Built production-grade Flask endpoints for uploading and classifying medical images, serving 10K+ monthly API calls from clinicians.
- Engineered access control and image streaming features to prevent downloads of sensitive clinical images, aligning with HIPAA compliance.
- Collaborated cross-functionally to standardize metadata schemas and optimize model deployment workflows.

VEYTEL, INC — Remote

Software Development Intern — Jun 2023 – Aug 2023

- Developed Python scripts to automate the renaming and reorganization of 25,000+ medical imaging files improving data accessibility and reducing internal search time by 40%.
- Collaborated in an agile team environment to build and debug production-level code, resulting in a 15% decrease in system errors.
- Delivered weekly updates and technical documentation, increasing project transparency and alignment across cross-functional teams.
- Utilized Google Colab to prototype and test scripts efficiently in a cloud-based environment (data pipelines and scalable tooling).
- Applied software engineering best practices in code reviews, and modular development to support digital asset management goals.

BARNES & NOBLE EDUCATION (BARTLEBY) — Remote

(A $1.6 billion Higher Education / Digital Education company with 4K employees; Bartleby was a high growth EdTech start-up within BNED)

Application Development Assistant — Mar 2021 – Jun 2021

- Conducted 400+ test runs on a Math native app for Bartleby verifying the accuracy and functionality across multiple modules.
- Authored original test cases by analyzing problem sets from 50+ textbooks, ensuring content alignment with educational standards.
- Prepared detailed QA reports and progress updates in collaboration with senior developers, supporting product quality and release.
- Participated in brainstorming sessions to propose enhancements to question variety, logic flow, and learner experience.
- Gained exposure to structured software testing, QA documentation, and content-driven product development in a real-world setting.

EDUCATION

MOUNT HOLYOKE COLLEGE — South Hadley, MA

B.A., Computer Science | Major GPA: 3.61/4.0 — May 2025

- **Relevant Coursework:** Data Structures, Operating Systems, Algorithms, Machine Learning, Applied Cryptography
- Scholarship recipient · Resident Advisor (2022–23)

FOUNTAINHEAD SCHOOL — Surat, India

High School Diploma — May 2021

International Baccalaureate Diploma (2021) | Final Score: 35/45 | HL: Mathematics AA, Economics, Physics

PROJECTS

ROBOTICS MOTION PLANNING ALGORITHM (C++, Linux VM) — April 2025

- Implemented a centroid-biased path planning algorithm to simulate autonomous robot movement through cluttered spaces, with added noise modeling to test robustness under uncertainty.

CLOUD BASED SENTIMENT ANALYSIS API (PYTHON + FLASK + GCP) — Nov 2024

- Designed and deployed a Flask-based REST API for real-time sentiment scoring on GCP using Docker containers. Implemented autoscaling and logging for monitoring and performance. Used in a demo web app with 1,000+ test queries.

FAKE NEWS CLASSIFIER (NLP) — May 2024

- Built a Naive Bayes NLP classifier for misinformation detection using 20K+ headlines. Tuned precision-recall tradeoffs and analyzed topic biases in social media datasets.